

Uncertainty-Aware Inference: Calibration Under Quantization

[Team A]

Ruchi Mahindru · mentor

Jiaming Liu · jl7250

Yanhao Bai · yb2630

Haotian Lei · hl3945

Leah Li · ql2481

GitHub: <https://github.com/chuanbinp/uncertainty-aware-inference>

Tracking: https://wandb.ai/Uncertainty_Aware_Inference_Lab/UAI_Project

Motivation & Background

Why calibration matters

- Guo et al. (2017) established ECE as the standard calibration metric, showing modern neural networks are systematically overconfident, meaning confidence rarely matches actual accuracy
- Kadavath et al. (2022) extended this to LLMs specifically. Even FP16 models are imperfectly calibrated, and calibration varies significantly across domains and task types
- Proskurina et al. (2024) showed GPTQ quantization increases calibration error across LLaMA and Mistral families, and yet quantization has become the default deployment method at scale

The Gap

Prior work is limited to a single PTQ method and a narrow set of models. Very few studies have done a cross-model, multi-PTQ calibration sweep, and to our knowledge, none have asked whether knowledge distillation can recover lost calibration quality.

Goal: Systematically study how NF4, GPTQ, and AWQ affect calibration across three model families, and whether temperature-scaled KD can recover it.

Research Questions & Scope

Research Questions

1. How does post-training quantization affect calibration?
2. Does KD help recover lost calibration?

Project Scope

- Three teams
- Three models families (Llama-1 7B, Mistral 7B, Llama-2 13B)
- Five model variants (FP16, GPTQ INT8/INT4, AWQ INT4, NF4)
- Three evaluation datasets (Hellaswag, TriviaQA, PubmedQA)

Team A's Role: LLama-1 7B

- Built the shared calibration pipeline (ECE, Brier Score, reliability diagrams) used by all three teams
- Evaluate all five variants of LLaMA-1 7B
- Built and ran the KD training and evaluation pipeline across single and dual temperature settings

Technical Approach — Model & Data

Model

Architectures: LLaMA-1 7B (TeamA Focus), Mistral 7B, Llama-2 13B.

Frameworks: PyTorch, HF Transformers, PEFT (LoRA), BitsAndBytes, AutoGPTQ, AutoAWQ.

Dataset

Evaluation:

- HellaSwag (Commonsense MC, 39.9k val split, MIT License)
- TriviaQA (Factoid QA, 11.3k val split, Apache 2.0)
- PubMedQA (Bio-medical QA, 500 test split, MIT License)

Distillation: C4 (English web text, ODC-BY License, subset of 2,000 training samples truncated to 1024 tokens) for KD training.

Hardware

1x RunPod A100 SXM4-80 GB GPU (CUDA 12.4 PyTorch 2.5).

Technical Approach — Calibration and Compression Applied

Instead of pure hardware optimizations, our optimizations focused on balancing footprint vs. prediction reliability.

1. Post Training Quantization (NF4, AWQ, GPTQ INT4/8)

Reduces memory footprint by ~70% using group-wise quantization and optimal data types, but distorts the output probability distribution.

2. Temperature-Scaled Knowledge Distillation

Training a 4-bit student (via LoRA) on soft targets from an FP16 teacher transfers the teacher's uncertainty and inter-class similarities.

3. Asymmetric Dual-Temperature ($T_t > T_s$)

Using a higher temperature for the teacher and lower for the student preserves calibration signals while preventing the student's distribution from collapsing to uniform noise.

Profiling Methodology

Focus on rigorous measurement and preventing misleading metrics.

Tools

- HF Transformers / PyTorch for standard generation logging.
- netcal package for rigorous ECE and MCE math.
- Weights & Biases for run-level tracking and dashboarding.
- Matplotlib custom scripts for Reliability and Entropy Distribution diagrams.

Metrics captured

- Calibration: Expected Calibration Error (ECE), Max Calibration Error (MCE), Brier Score, Average Entropy.
- Task Quality: Accuracy (Zero-shot, Log-likelihood choice scoring, and Greedy generative QA).
- Systems/Hardware: GPU Memory Footprint (GB) and Throughput (Tokens/sec).

Ensuring measurement stability:

- Consistency: Fixed random seeds across all scripts and standardized generation parameters.
- Isolation: All latency and throughput profiling runs were isolated on a single A100 to prevent multi-tenant interference.

Results — PTQ Results

TABLE II
PTQ ACCURACY AND ECE ACROSS MODELS, QUANTIZATION METHODS, AND DATASETS. BEST ECE PER MODEL-DATASET IN **BOLD**.

Model	Method	HellaSwag		TriviaQA		PubmedQA	
		Acc $\uparrow$	ECE $\downarrow$	Acc $\uparrow$	ECE $\downarrow$	Acc $\uparrow$	ECE $\downarrow$
LLaMA-1 7B	FP16	0.7395	0.3086	0.6854	0.0837	0.5360	0.2392
	GPTQ-INT8	0.7394	0.3085	0.6860	0.0823	0.5350	0.2310
	GPTQ-INT4	0.7300	0.3015	0.6494	0.0763	0.5390	0.2402
	AWQ-INT4	0.7328	0.3040	0.6752	0.0707	0.5390	0.2552
	NF4	0.7342	0.3021	0.6616	0.0750	0.5320	0.2469
Mistral 7B	FP16	0.8293	0.2771	0.7005	0.1427	0.4910	0.2293
	GPTQ-INT8	0.8191	0.2827	0.6778	0.1550	0.4610	0.1033
	GPTQ-INT4	0.8246	0.2893	0.6807	0.1525	0.4930	0.0766
	AWQ-INT4	0.8192	0.2825	0.6854	0.1545	0.4920	0.0667
	NF4	0.8259	0.2793	0.6870	0.1581	0.4810	0.2361
Llama-2 13B	FP16	0.7750	0.3350	0.7670	0.1667	0.4700	0.2293
	GPTQ-INT8	0.7750	0.3349	0.7630	0.1633	0.4730	0.2284
	GPTQ-INT4	0.7690	0.3326	0.7430	0.1549	0.4730	0.2197
	AWQ-INT4	0.7750	0.3368	0.7520	0.1647	0.4710	0.2415
	NF4	0.7670	0.3298	0.7540	0.1775	0.4670	0.2400

quantization preserves ECE

Results — KD Temperature Sweep

TABLE V
KD TEMPERATURE SWEEP — LLAMA-2 13B. PUBMEDQA ECE
IMPROVES MONOTONICALLY.

T	HellaSwag			TriviaQA			PubmedQA		
	Acc	ECE	Brier	Acc	ECE	Brier	Acc	ECE	Brier
NF4	.7670	.3298	.2725	.7540	.1775	.2241	.4670	.2400	.3097
1	.7500	.3129	.2684	.7360	.1615	.2263	.4350	.2409	.3196
2	.7660	.3287	.2717	.7390	.1605	.2256	.4750	.2352	.3071
4	.7660	.3297	.2730	.7450	.1754	.2253	.4740	.2269	.3043
8	.7590	.3233	.2709	.7440	.1691	.2230	.4750	.2163	.2991
12	.7610	.3254	.2717	.7460	.1730	.2230	.4800	.2139	.3008
16	.7590	.3240	.2711	.7450	.1654	.2196	.4720	.2204	.2998

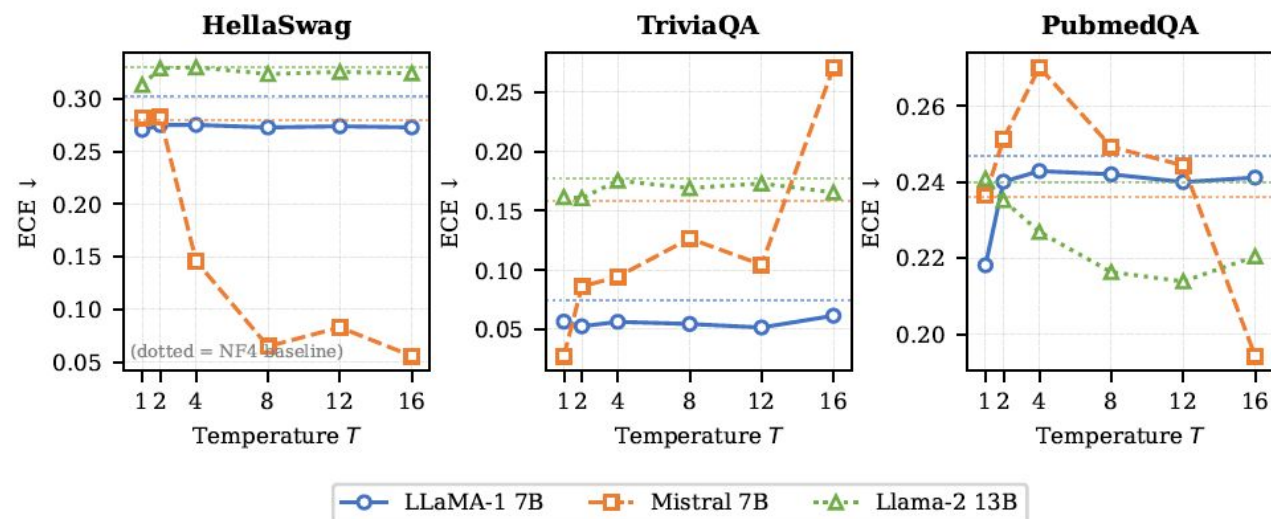


Fig. 3. ECE vs. KD temperature for all three models on HellaSwag (left), TriviaQA (center), and PubmedQA (right). Dotted lines indicate NF4 baseline ECE. Mistral’s ECE drop at $T \geq 4$ is an artifact of distribution collapse (§V-C). Llama-2’s monotonic decrease on PubmedQA is genuine (§V-D).

Three models, three behaviors: LLaMA-1 is flat, Mistral’s ECE drops are an artifact of collapse at $T \geq 4$, and Llama-2 13B shows a genuine monotonic ECE improvement on PubMedQA (0.240 $\rightarrow$ 0.214) with stable accuracy.

Results — Why ECE Lies

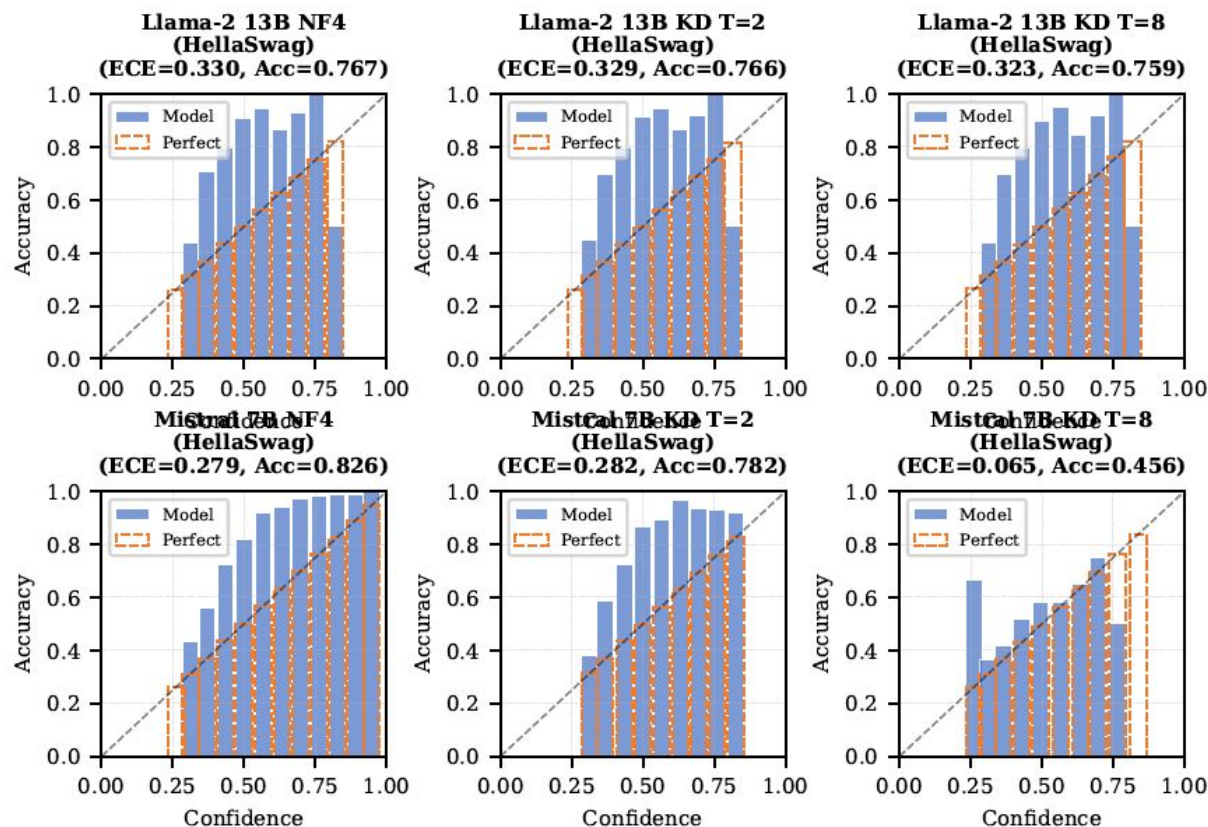


Fig. 5. Reliability diagrams (HellaSwag, 1,000 samples) at NF4 baseline, KD $T=2$, and KD $T=8$. Top: Llama-2 13B. Bottom: Mistral 7B. Mistral at $T=8$ shows near-uniform confidence (degenerate); Llama-2 remains structured.

Lessons Learned

What worked

- Shared calibration pipeline worked: one eval flow produced confidence, ECE/MCE, Brier, entropy, reliability plots, and raw tensors.
- NF4 was the practical KD student format: QLoRA/bitsandbytes supported LoRA backprop cleanly in our setup.
- ECE alone was not enough: Brier + entropy exposed near-uniform collapse.

What didn't work

- High-temperature KD broke Mistral: entropy rose, outputs became near-uniform, and HellaSwag accuracy fell from 82.6% to 43.4%.
- Dual-temperature KD did not rescue Mistral: even $T_t=2$, $T_s=1$ dropped HellaSwag accuracy to 51.5%.
- Generic C4 KD did not preserve task accuracy everywhere: LLaMA-1 TriviaQA lost 8–9pp under single-temperature KD.

Next Steps

- Task-aligned KD data: train on PubMed abstracts or TriviaQA passages instead of C4 to test whether accuracy can recover while preserving calibration gains.
- QAT vs. PTQ on LLaMA-1 7B: evaluate whether quantization-aware training preserves calibration better than post-hoc KD.
- Routing with calibration safeguards: test NF4-first / FP16-fallback routing, but include Brier or entropy checks to avoid confidence failures.

Teamwork & Contributions

Member	Primary contributions	Tools / artifacts
Jiaming Liu	Modularized calibration evaluation pipeline; shared model loading and dataset modules used by all three teams; ran evaluation for LLaMA-1 7B FP16, AWQ-INT4, and NF4 baselines; dual-temperature KD training and evaluation	Part of shared/eval_utils.py, shared/model_loader.py, shared/data_loader.py, TeamA/run_eval.py, TeamA/run_eval_args.py, FP16, AQW INT4 and NF4 evaluation results, KD dual-temp results
Yanhao Bai	Setup calibration evaluation groundings; implemented ECE, Brier, reliability diagrams used by all three teams; ran evaluation for LLaMA-1 7B GPTQ-INT8, GPTQ-INT4	TeamA/colablink.md, Part of shared/eval_utils.py, GPTQ INT8 and INT4 evaluation results
Haotian Lei	Led Team A report integration and KD result analysis; validated single- and dual-temperature KD results; analyzed ECE/Brier/entropy failure modes and accuracy-calibration tradeoffs;	TeamA/kd_train.py, TeamA/run_eval_kd.py, TeamA/results/kd/single-temp KD results
Leah Li	Ran calibration evaluation across Llama-2 7B, Mistral 7B, and Llama-2 13B (FP16, LLM.int8, GPTQ-INT4, AWQ-INT4, NF4, NF4+KD); ran KD single-temperature sweep; produced headline results table; contributed to report writing and figures	Llama-2 7B, Mistral 7B, Llama-2 13B FP16 / LLM.int8 / GPTQ-INT4 / AWQ-INT4 / NF4 / NF4+KD evaluation results, single-temp KD sweep results

Collaboration: regular [weekly / twice-weekly] meetings via [Zoom / in person], shared [Slack / Discord] channel, code review on every PR.

Thank you

Questions?

GitHub: <https://github.com/chuanbinp/uncertainty-aware-inference>

Tracking: https://wandb.ai/Uncertainty_Aware_Inference_Lab/UAI_Project

Backup — Team Structure

Backup slide. Use for Q&A. Add more backup slides as needed.

Team	Model	Cross-Cutting Specialty	Shared Artifact Owned
Team A	Llama-2 7B	Builds shared calibration pipeline, leads KD experiments, owns QAT stretch goal	Calibration evaluation framework (ECE, Brier, reliability diagrams)
Team B	Mistral 7B	Leads CUDA/Nsight profiling, Roofline analysis, sets up vLLM serving infrastructure	Profiling harness + vLLM serving setup
Team C	Llama-2 13B	Leads cross-model Pareto analysis, routing simulation, report structure, final visualizations	Pareto dashboard + routing simulation + report template